Relighter: Ground Truth Free Learned Single Image Relighting

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Abstract

We show how to relight a scene, depicted in a single image, such that (a) the overall shading has changed and (b) the resulting image looks like a natural image of that scene. Applications for such a procedure include generating training data and building authoring environments. Naive methods for doing this fail, because shading and albedo are quite strongly related; for example, sharp boundaries in shading tend to appear at depth discontinuities, which are usually apparent in albedo. Because the same scene can be lit in different ways, our method produces a representation of the available lighting fields in the form of imputed generators of the illumination cone. Our procedure does not require expensive "inverse graphics" datasets, and sees no ground truth data of any kind, but instead exploits an adversarial loss. Our network accepts an image, and its albedo map, and produces a representation of the generators of the illumination cone. The network is trained by requiring that shading the albedo field with a random convex combination of generators should result in an image that is indistinguishable from real images. We offer a summary quantitative evaluation of the method with a novel application of the FID. An extension of the FID allows per-generated-image evaluation. Furthermore, we offer qualitative evaluation with a user study.

1 Introduction

When someone switches on a light, the image at the back of your eye changes even when there is no other change. The new shading of the scene in front of you is determined by the shape and material composition of the scene, and by the light source. This paper shows how to **relight** a scene – we demonstrate a model that can generate images of the scene with random (but plausible) changes in light source. One might expect scene relighting to be easy – decompose the scene into albedo and shading, and replace the shading with another shading field. This approach doesn't work because shading fields are very strongly related to albedo fields and to shape. For example, sharp boundaries in shading tend to appear at depth discontinuities, which usually apparent in albedo. CycleGAN [62] isn't appropriate, as the relit image must be an image of the original scene. This means that it is hard to produce shading fields that apply to a particular scene with elementary methods.

An alternative strategy is to use "inverse graphics" – learn to regress geometry, material, etc. against an image using training set that consists of ground truth geometry, material etc. together with computer generated images (CGI). But, while this strategy may offer an approach to building some kinds of vision system, it is laborious (one must produce the training data) and is not yet reliable. Furthermore, the inverse graphics strategy is intellectually unsatisfying, because it doesn't explain how a visual agent learns to interpret the rich visual world they see if the agent doesn't have a CGI training set conveniently to hand.

Instead, we see scene relighting as an unsupervised problem. We apply an intrinsic image method (supervised only with abstract spatial models) to a collection of scene images, and then model the

Preprint. Under review.



Figure 1: **Top:** *True images of scenes.* **Other rows:** *Relighted images of those scenes. Scenes are the first 8 scenes of the bedroom category of the MIT Places2 dataset; there is no curation.*

dependency of shading on albedo by inferring a set of illumination cone generators (ICG). Any non-negative combination of these generators is a physically plausible shading field. The trick is inferring these generators without having multiple shadings of the same scene; but scenes have highly stylized shapes [50], and scenes with similar geometry should have similar generators. In turn, we can obtain generators by looking at a large, unlabelled dataset. Our process uses physical and spatial models, but requires no data marked up in any form.

Relighting methods should offer: **diversity**, so the method should be able to produce multiple different lightings of a scene; **realism**, so relightings look like plausible images of that scene; **extrapolation**, so some relighted images should represent lighting conditions that might not appear in a reference set of (say) bedroom images (for example, very deep shadows; strongly directional lighting); and **control**, so one can require the method to produce particular relightings of a scene.

Contributions: We offer the first scene relighting method that requires no data marked up in any form. We show how to evaluate our models quantitatively as well as qualitatively. We introduce a variant of the FID that applies to image transformations and gives a per-image estimate of quality. Our methods offer strong diversity and realism, limited extrapolation and weak control (section 6).

2 Related Work

Intrinsic image decomposition Decomposition into albedo and shading is well established. Land's influential Retinex model assumes effective albedo displays sharp, localized changes (which result in large image gradients), and that shading has small gradients. Land produced numerous variants [32, 33, 35, 34], and the computer vision community has produced a number of variants under the same name [22, 23, 46, 6, 9, 10, 8, 19, 56, 29, 16, 39, 52, 38, 18, 7, 59]. No ground truth is required by these models. The recent literature strongly emphasizes directly supervised CNN based models, using IIW data (data [4]; methods [48, 61]) or CGI data [11, 48, 40]. Competitive modern methods are trained using a training portion of the IIW dataset, then evaluated on a the test portion. [17] obtain a SOTA WHDR of 14.45% in this way, but their method produces strange colors in albedo images, making its applicability in computational photography questionable and qualitative comparison unhelpful. [5] use a similar approach, but different network architectures, to obtain a mean WHDR of 17.18% with strong qualitative results.There is good evidence that relatively little supervision is required, and that

self-supervision can be successful [24, 12]. Some methods not see the albedo and shading of training images, but instead equivalent information, such as aligned views of the same scene under distinct illuminants (real [56, 41, 30]; CGI [37]; partial alignment [58]) or only statistical models of albedo (resp. shading). [45] obtains the current SOTA WHDR for methods that use only indirect supervision (18.69%)

Equivariance: A function $\phi : \mathbf{x} \in X \rightarrow \mathbf{y} \in Y$ is equivariant under the action of a group G if there are actions of G on X and Y such that $\phi(g \circ \mathbf{x}) = g \circ \phi(\mathbf{x})$. Information being gained or lost at the boundary is an obstacle to applying the theory of group actions exactly (except for certain finite groups [14]). Well-known visual feature representations tend to have good equivariance-like properties either by design or in practice [36]. Generally, equivariance properties have not been imposed on regression networks.

Inverse rendering seeks to recover all required data for rendering from an image. Reconstruction methods applied to multiple images are one approach (eg [28]); another is to impose strong parametric models [2]; alternatively, one could train a regressor using a fixed object class (faces [53]; furniture [54]; birds [25]) or multiple images [58]. Very strong reconstructions can be obtained from rendered images [42].

Object Insertion starts with Lalonde *et. al.* [31], who insert fragments into target images. Karsch et al show that computer graphics objects can be convincingly inserted into inverse rendering models obtained with geometric inference [26] or with single image depth reconstruction [27]. Inverse rendering trained with rendered images can produce excellent reshading of computer graphics objects [51]. However, recovering a renderable model from an image fragment is extremely difficult, particularly if the fragment has odd surface texture. Liao *et al.* showed that a weak geometric model of the fragment can be sufficient to correct, if one has strong geometric information about the target scene [43, 44]. In contrast, our work is entirely image based: we take a scene, and relight it.

Illumination cone: The family of images created by lighting a single convex diffuse object, viewed in a fixed camera, with a point source is naturally a convex cone (reflection is linear; irradiance is non-negative). As Kriegman and Belhumeur [3] point out, this fact extends to any diffuse scene. If the observed scene has only a discrete set of normals – for example, a polyhedral object – then the cone is polyhedral and has a finite set of generators (i.e. any element of the cone is a non-negative sum of a finite set of generators). There is a straightforward procedure for recovering generators from multiple images in simple cases. Interreflection and shadowing may mean that the cone is not polyhedral, but experiments suggest that (a) relatively few generators provide a good representation of the cone anyhow ([3], p.8; [20], p.644) (b) the cone is relatively "flat" ([3], conjecture 1; p.12;) and (c) illumination cone methods offer strong models of extreme relighting for face images [20].

2.1 Intrinsic Image Decomposition

Our intrinsic image decomposition method is described in detail in [1], currently in review (full anonymized version in supplementary). A summary appears here for the readers' convenience. We wish to predict albedo and shading fields ($\mathcal{A}$ and $\mathcal{S}$) from an image $\mathcal{I}$, but have no examples where ground truth is known. However, intrinsic images have several important properties. An intrinsic image decomposition should **explain** the image – pixel values should be accurately predicted by albedo and shading. Intrinsic images have a **local character** – one can tell whether a moderately sized image patch is an albedo (resp. shading) patch without reference to the rest of the albedo (resp. shading) field. Finally, the mapping from an image to an intrinsic image is **equivariant** under image translation and orthonormal transformations – for any two images of the same scene, the albedo (resp. shading) reported for overlapping regions should be the same. Similarly, the mapping from an image to an intrinsic image is somewhat scale equivariant.

We train a network to decompose fixed size *tiles* (128×128) to albedo and shading estimates. The local character of the problem means we can train a network using synthetic albedo (resp. shading) fields that need be accurate models only locally. We use straightforward statistical models of albedo and shading (**paradigms**) that (a) capture the main local features of albedo and shading and (b) are easy to sample. Our albedo paradigm captures: albedoes are piecewise constant; the color distribution should reflect likely surface colors; there should be a profusion of edges with no strong orientation bias; there should be at least some vertices with degree greater than three. Similarly, our shading

paradigm captures: shading contains many slow and very slow changes; there are some sharp shading edges; and the dynamic range of shading indoors is limited.

The network is trained using two kinds of tile. **Fake tiles:** are the product of samples from albedo and shading paradigms; the method is trained to decompose the product into the original samples. **Real tiles:** are tiles from real images; the method must produce albedo and shading that multiply to approximately the original tile, and the reported albedo (resp. shading) fields should "look like" the synthetic fields locally, too. We achieve this similarity with an adversarial loss. This supplies an *adversarial smoother* – the decomposer cannot report *anything* for real images, but must confine itself to outputs that are "like" the synthetic fields.

Equivariance by averaging: To ensure that we report a translation equivariant estimate, we cover a real image with randomly offset, overlapping tiles, compute albedo (resp. shading) fields for the tiles, then average, so the albedo at a given location is an average over all tiles covering that location. We apply this procedure over several scales and average the result to obtain scale equivariance. We compute a scale and translation equivariant estimate for a discrete set of rotations and reflections, and average those to estimate a rotation equivariant result. Finally, a simple pointwise procedure ensures that the residual is small. The resulting method is significantly better at the standard evaluation (WHDR [4], computed on the IIW images of that paper) than SOTA, for methods that see no IIW data in training (compare our 17.06% with 18.69% [45] and 19.5% for Retinex [49]).

3 Relighting Scenes

Our model maps an image $I(c)$ of a given scene c and an estimate of scene albedo to a set of illumination cone generators (ICG) $g_i(c)$. We assume that a fixed number N_g of generators is sufficient to model the shading. Furthermore, we will assume that the generators are scaled such that a convex combination of the generators is a natural image of the scene; equivalently, the camera scaling constant is such that we always observe a convex combination of generators.

The model needs to see $I(c)$, because $I(c)$ contains structural information about the scene (depth, normals, etc. can be extracted from $I(c)$ as required, eg [55, 15]), suggesting that generators can be extracted from $I(c)$. Furthermore, the model needs to see an albedo estimate, because a generator could be very bright in locations with very dark albedo without causing the image pixels to be too bright. Regressing structure against $I(c)$, then imputing generators from that structure, presents difficulties: first, regressing structure requires large quantities of labelled data; second, imputing generators from structure requires knowing at least a prior model in illumination fields. But all this information should be present in large pools of images of given type of scene, so we seek to construct generators directly. We model ICGs with a network $\left[g_1, \ldots, g_{N_g}\right] = N(I, A; \theta)$. The lighting in any particular image is a non-negative combination of the ICG. We model the weights as distributed according to a symmetric Dirichlet distribution with concentration parameter α (choice of α discussed below). Then we can generate relighted images by: computing a weighted combination of generators; and multiplying by albedo. An adversary then tests whether the result is realistic.

This test is necessary, but not sufficient. Networks have a strong tendency to cheat; in this case, natural cheating strategies are to report ICGs that: yield only the original shading; are not linearly independent; or do not yield the original shading. We use a set of losses, below, to prevent this cheating.

3.1 System Details

We wish to model possible shading fields $S(c)$ for an image $I(c)$ of a given scene c, using generators $g_i(c)$ of the scene's illumination cone. We will assume that a fixed number N_g of generators is sufficient to model the shading, and the camera scaling constant is such that we always observe a convex combination of generators. We modelling generators with a network $\left[g_1, \ldots, g_{N_g}\right] = N(I, A; \theta)$. We implement $N(I, A; \theta)$ using a u-net, with skip connections (details in supplementary; θ parameters). More detail in supplementary material.

Losses: The key difficulty here is that we do not actually know the generators, or the appropriate combination of generators to produce any observed shading field. However, we can construct relighted images from an image of a scene by: computing the generators; constructing a shading field (i.e. a

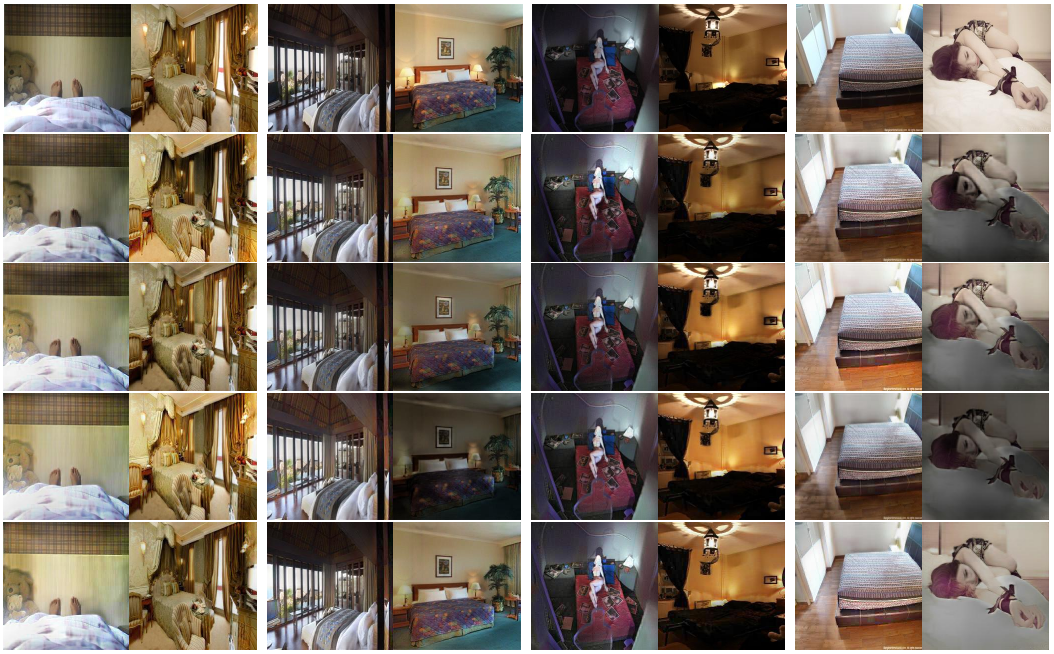


Figure 2: **Top row:** *real images of scenes from the bedroom category.* **Other rows:** *relighted images of those scenes. Each pair of columns shows a set of relighted versions randomly chosen in the corresponding quartile of mean LFID over four random reshadings (so first column has low mean LFID, etc)*

weighted sum of generators, with non-negative weights); and then multiplying by the albedo A. Write

$$R(c; g(I, A), w, A, \theta) = (\sum_i w_i g_i(I(c), A, \theta))A$$

for such a relighted image. To obtain diverse shading fields, we must ensure the generators for a given scene are distinct from one another. Furthermore, any relighted image must be indistinguishable from a real image.

Generator spread: We compute a smoothed version of each generator $s_i = G_\sigma * g_i$, normalize to obtain $n_i = s_i / \| s_i \|_2$, then compute $\mathcal{L}_d(g) = -\log\det N$, where the i, j'th component of N is $n_i^T n_j$.

Decorrelation with lightness: there is no reason that the shading generators should be correlated to albedo, so we require each individual generator is decorrelated from the surface lightness at a long scale. This helps avoid generators that simply make dark objects darker and light objects lighter. Writing $l(c)$ for the lightness (average of r, g, and b albedo), we compute

$$\mathcal{L}_c(g) = \sum_i \left[\text{corr}(G_\sigma * g_i, G_\sigma * l)\right]^2$$

Realism housekeeping: we require that any pixel in R should be non-negative ($\mathcal{L}_u(R)$), and less than one ($\mathcal{L}_o(R)$), and achieve this with simple one-sided quadratic losses. Note that this constraint applies to relighted images, rather than shading fields – applying the constraint to shading fields will lead to regions with dark albedo being consistently dark. We show the network A as well as I so that it can anticipate this loss and produce very bright shading at locations where the albedo is dark.

Consistency: the original image of the scene should be in the span of the generators, so there is some $w = \hat{w}$ such that the original image of the scene can be recovered from the generators (i.e. $R(c; g, \hat{w}, A, \theta) = I(c)$). We use $\mathcal{L}_s(R) = \| R(c; g, \hat{w}, A, \theta) - I(c) \|_2^2$, where $\hat{w}$ is a linear least squares estimate. We impose the requirement that $\hat{w}$ is non-negative using a one-sided quadratic loss ($\mathcal{L}_{w,u}(\hat{w})$).

Realism from a discriminator: relighted images should be realistic. We impose an adversarial loss $\mathcal{L}_a$ – relighted images should be indistinguishable from real images. We ensure that the real images

used to train the adversary are distinct from the images used to train the network, to prevent the method using the adversary to cheat.

Overall loss: the loss is then:

$$\mathcal{L} = \lambda_d \mathcal{L}_d + \lambda_c \mathcal{L}_c + \lambda_u \mathcal{L}_u + \lambda_o \mathcal{L}_o + \lambda_s \mathcal{L}_s + \lambda_{w,u} \mathcal{L}_{w,u} + \lambda_a \mathcal{L}_a$$

3.2 Inference and Equivariance

Our network is trained on 128x128 images, but is applied to 256x256 images. Relighting has a form of equivariance property: small movements of the camera frame should not affect the overall relighting of a scene (apart from rows and columns of the image moving in and out of view). We have not been able successfully to impose this property by losses, and so we compute relighting effects on multiple tiles, then average. Assume we know an image, albedo, and set of generator weights for inference. We rescale the image (resp. albedo) to 160x160, then reflection pad rows and columns to 200x200; this allows tiles to have support outside the original image. We then compute relightings for each of 3 downsampled versions of the new image/albedo pair, then upsample and average. Each relighting is computed by covering the new version with a 7x7 grid of overlapping tiles of dimension 128x128, with jittered centers, and computing a relighting of that tile; the results are averaged. The choice of dimensions, number of scales, and number of translations is arbitrary, but constrained by the requirement that a relighter should see "most" of a scene to be able to produce a reasonable prediction. Averaging improves results (supplementary), but we have no information about the best choice of these hyperparameters.

4 Quantitative Evaluation for Relighting

We need quantitative measures. An important difference between image relighting and image generation is that relighting should map the collection of real images to itself, preserving frequencies etc. (loosely, relighting is an *isomorphism*). This yields both a measure of population quality and one of per-image quality.

4.1 Measures

Realism in current generators is typically measured with the FID [21]. One embeds a set of generated images, then computes the Frechét inception distance between the resulting points and an embedding of a set of real images. We use the standard VGG embedding of [21], with dimension $d_f = 2048$. This is an estimate of the difference between the distributions from which the samples were taken; smaller values are better. As [13] show, the FID estimated from a finite set of points is biased by a $1/N$ term that depends on details of the generator, but straightforward extrapolation methods remove this bias successfully to produce FID_∞. Because relighting is an isomorphism, the FID between a set of images $\mathcal{O}$ and a relighted set $\mathcal{R}$ of images should be zero, so $\text{FID}_\infty(\mathcal{O}, \mathcal{R})$ should be zero up to sampling error.

Diversity: We must also evaluate whether relighted versions of an image are actually different from the original image. We measure **diversity** by comparing relighted images to the original; write N_t for the total number of pixels in relighted images and compute MSD where

$$MSD_{\hat{\theta}} = \frac{1}{N_t} \sum_{i \in \text{images}} \sum_{k \in \text{pixels}} (I_{i,k} - R_{i,k})^2$$

Note quite unreasonable shadings may have high diversity scores.

Baselines: There are no meaningfully comparable methods. However, sampling error, etc., means it is important to have baselines for "low" and "high" values. We use three baselines. **Spl:** we evaluate sampling variance in the FID by computing the FID between random splits $\mathcal{O}_i$ of the image set, and obtain $F_{\text{spl}} = \text{FID}_\infty(\mathcal{O}_i, \mathcal{O}_j)$. **Iid**: we evaluate the effect of the intrinsic image decomposer by comparing images with the albedo-shading products $\mathcal{P}$, obtaining $F_{\text{iid}} = \text{FID}_\infty(\mathcal{P}, \mathcal{O})$. Each should have "low" FID_∞. **Shf**: we decompose images, shuffle the shading fields, and recompose to obtain $\mathcal{P}_s$. We then compare to originals, obtaining $F_{\text{shf}} = \text{FID}_\infty(\mathcal{P}_s, \mathcal{O})$. The **Iid** baseline yields a "low" MSD estimate, and the **Shf** baseline should establish a "high" MSD estimate.

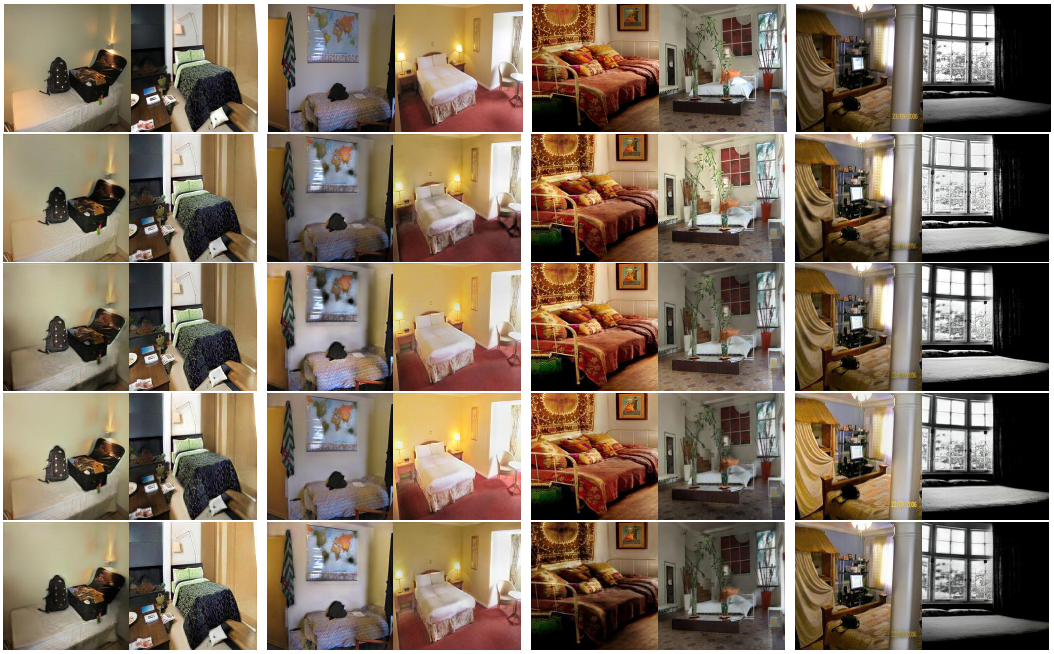


Figure 3: **Top row:** *real images of scenes from the bedroom category.* **Other rows:** *relighted images of those scenes. Each pair of columns shows a set of relighted versions randomly chosen in the corresponding quartile of mean MSD over four random reshadings (so first column has low mean MSD, etc)*

4.2 The Local FID

Because relighting is an isomorphism, we can recover a per-image measure of quality. Assume we obtain $\mathcal{R}$ from $\mathcal{O}$ (a set of N images) by relighting a single image I_k and compute FID$(\mathcal{O}, \mathcal{R})$. This value – which is non-negative, and is ideally zero – measures the extent to which the relighted image is realistic. Furthermore, different relightings of the same image will have different values. However, computing this value requires care, because simply substituting and recomputing results in numerical problems. Instead, we look for an assymptotic estimate. Writing μ_O for the mean and C_O for the covariance of $\mathcal{O}$ in the chosen embedding, rotated so that C_O is diagonal; write $\mathbf{x}(I_k)$ for the embedding of I_k, R_k for the result of relighting I_k, and $\mathbf{d}_k = \mathbf{x}(R_k) - \mathbf{x}(I_k)$. As supplementary material demonstrates, we have FID$(\mathcal{O}, \mathcal{R}) = (1/N^2)(\mathbf{d}_k^T \mathbf{d}_k + \text{Trace}(C_O^{-1} M_1^2)) + O((1/N^3))$. (where M_1 can be evaluated in a straightforward fashion, supplementary). We define the **local FID** at image k to be:

$$\text{FID}_{l,\mathcal{O}}(I_k) = (\mathbf{d}_k^T \mathbf{d}_k + \text{Trace}(C_O^{-1} M_1^2))$$

This construction can be applied to any method that ostensibly transforms a real image into another real image.

4.3 Experimental Setup

Datasets: We train with images from the LSUN Bedroom dataset ([57]), resized to 128x128. Training albedos are obtained with the method of section 2.1. We evaluate with training images from the bedroom category of the MIT Places dataset [60]. Images are decomposed into albedo and shading at 256 x 256 resolution using the method of section 2.1 (which sees no ground truth). We evaluate generalization with other rooms from that dataset (bathroom;childs room;dining room;dressing room;living room).

Dirichlet parameter: A small value of α_{train} means that relatively few generators participate in any particular sample, and so there is less opportunity for error terms to cancel. A large value presents the risk that there may be unconstrained components of error that cancel across generators. Using an α_{test} that is smaller than α_{train} forces the method to extrapolate.

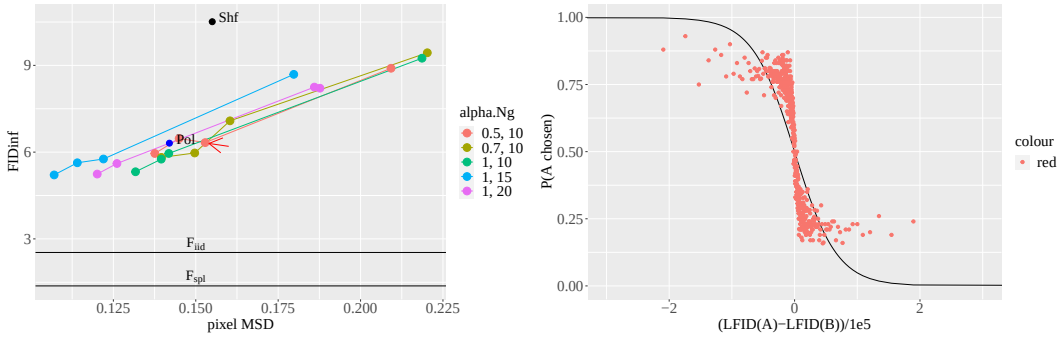


Figure 4: **Left:** *FID against pixel mean-squared difference for 5000 bedroom images from the MIT PlacesRooms dataset, for various models. Each curve represents a choice of α_{train} and N_g (legend to right). Markers represent observed values for $\alpha_{test} \in \{1, 0.7, 0.5, 0.1\}$. Increasing values of $\alpha_{test} \in (0.1, 0.5, 0.7, 1)$ result in models with larger MSD and larger FID. The Pareto frontier is down and to the right; settings on that frontier represent a compromise between realism and diversity.* **arrow***: Model chosen for polishing, $(\alpha_{train}, N_g, \alpha_{test}) = (1, 15, .5)$.* $F_{\mathbf{spl}}$*: FID_∞ obtained from random splits of dataset.* $F_{\mathbf{iid}}$*: FID_∞ obtained decomposing, then recomposing images (MSD is 0.03, and not shown because it rescales the plot).* **Shf***: MSD and FID_∞ obtained by a random shuffle of shading fields.* **Pol***: MSD and FID_∞ obtained by polished version of chosen model.* **Right:** *Difference in LFID is a good predictor of human subject preferences. Points show probability that A is selected against difference value, for each of 500 quantiles of LFID difference and a total of 50000 preference tests. Curve shows a logistic regression predictor; the fit would likely be improved by removing large differences in LFID. Choice of scale has occluded two data points (one at very far right; one at very far left).*

Choice of hyperparameters: Our discriminator uses a hinge loss and is spectrally normalized (as in [47]). For model search purposes, we train the generator with 30K LSUN Bedroom images and their albedoes, and the discriminator with 30K distinct images (to prevent shading "leaking"). We fix scale by $\lambda_a = 1$. Simple qualitative experiments suggest the model is not sensitive to the choice of $(\lambda_d, \lambda_c, \lambda_u, \lambda_o, \lambda_s, \lambda_{w,u})$, and we use $(0.1, 1, 50, 50, 100, 50)$. Hyperparameters that have significant effects are: N_g, α_{train},α_{test} and discriminator. Figure 4 plots FID_∞ against MSD for a variety of methods. All methods achieve much larger MSD than the IID baseline; all methods have notably higher FID_∞ than F_{spl}. For each model, choice of α_{test} offers a way to pay off MSD against FID (larger α_{test} means more ICGs contribute to shading and produces lower MSD and FID). We choose the model indicated with an arrow for strong MSD with low FID, and then retrain with 100K LSUN Bedroom images for the generator and a different 100K for the discriminator. Polishing weakens MSD for minimal change in FID; we do not know why. A variant of the polished model is used to calibrate LFID; the polished model is used for all qualitative results.

5 Results

Qualitative analysis: Figure 1 compares four random relightings each with 8 images of scenes (scenes are curated; relighting is random). Figure 2 shows four random relightings each from the four quartiles of LFID averaged over each images relighting (showing relightings with small, etc. range of LFID). Figure 3 shows four random relightings each from the four quartiles of LFID averaged over each images relighting (showing relightings with small, etc. range of MSD). The model is clearly capable of substantial, non-obvious relightings that look realistic, and can clearly cope with a very diverse range of surfaces. There are both overall changes in brightness, and subtle local effects. The model appears to have some form of geometric knowledge (it often changes the lighting of horizontal and vertical surfaces differently).

Figure 4 shows that **LFID is a good predictor of human preference**. We show human subjects a pair of images (A, B) *of the same scene*, and record which is regarded as more realistic. Images are chosen uniformly at random from the original, and four relightings. We have 5000 runs of 10 choices, with anonymous subjects. The scatter plot shows frequency that A is preferred against LFID(A)-LFID(B) for 500 quantiles of the difference (so frequencies are estimated with 100 points).



Figure 5: **Left column:** *the first three scenes in the bedroom category;* **center three columns:** *for second, third and fourth quartile of MSD, the relighted image with smallest LFID chosen from 50 random samples (so third column is lowest FID with rather large MSD, etc.);* **right three columns:** *for second, third and fourth quartile of LFID, the relighted image with largest MSD chosen from 50 random samples (so first column is most distant image with rather small LFID, etc.)*

	bedroom	bath	childs	dining	dressing	living
mean LFID	31205	32691	38546	24970	42169	29011
mean MSD	0.133	0.136	0.152	0.124	0.127	0.126

Table 1: *Our method appears robust to changes of image category; performance as measured by LFID and MSD for the method applied to other rooms in Places 2 (bedroom is a training category, though training is on LSUN bedrooms).*

Subjects reliably choose the image with lower LFID, though probabilities at large differences saturate. The variation in frequencies suggest that other predictors might be required to score realism of an image transformation more accurately.

The uses of LFID: If relighting is to be used for data augmentation, one seeks relightings that make large changes to a scene. Figure 5 shows the results of a rejection sampling method that seeks the best LFID for MSD over a threshold; and the largest MSD for LFID below a threshold.

Relighting generalization: The training category for our method was bedroom, though training is on LSUN bedroom images and testing on Places 2 images. Our method appears to generalize across at least some scene categories. We evaluate mean LFID and mean MSD for the first five non-bedroom categories in Places 2 (training image set, which is larger) in table 1. Note that the mean LFID and mean MSD is not massively affected by choice of dataset.

6 Discussion

Limitations: While the model can achieve moderate relightings, extreme effects – for example, novel cast shadows or strongly directional illumination – are not present; and ringing occurs at some albedo edges. We have investigated controlling relighting (by choosing non-negative weights that produce a shading "close" to the demand); the resulting images are of variable quality and fidelity (supplementary). We have no clear procedure for improving the model, save collecting more (unlabelled) data.

Negative social implications: Convincing fake images can pollute (say) the news media; but our method offers only relighting, rather than the much more dangerous insertion or deletion editing. We have no example of a situation where a relighting procedure creates social harm. Experiments required approx 50 hours of time on a single GTX3090 from method inception to submission. The datasets used either do not contain, or are not known to contain, personally identifiable or offensive material. A total of $150 was paid to Amazon mechanical Turk workers for image labelling.

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Checklist

1. For all authors...
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]
   (b) Did you describe the limitations of your work? [Yes]
   (c) Did you discuss any potential negative societal impacts of your work? [Yes]
   (d) Have you read the ethics review guidelines and ensured that your paper conforms to them? [Yes]
2. If you are including theoretical results...
   (a) Did you state the full set of assumptions of all theoretical results? [Yes] LFID derived in supplementary
   (b) Did you include complete proofs of all theoretical results? [Yes] LFID derived in supplementary
3. If you ran experiments...
   (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? [No] Code is in a messy state, but will be available when polished; the text description is sufficient to reproduce results.
   (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? [Yes] Unsupervised
   (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? [N/A]
   (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets...
   (a) If your work uses existing assets, did you cite the creators? [Yes]
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   (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? [No] Material has been out for a long time, published by other researchers
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5. If you used crowdsourcing or conducted research with human subjects...
   (a) Did you include the full text of instructions given to participants and screenshots, if applicable? [Yes] In supplementary
   (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals, if applicable? [Yes] In supplementary; no risks
   (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [Yes] In supplementary; total of 15